



CAN DEBUG TOOL USER GUIDE



Politecnico
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OVERVIEW, SUPPORT AND INSTALLATION



Two main modes:

- **Receive Mode:** Monitors and decodes CAN messages in real time
- **Send Mode:** Sends CAN messages with precise frequency and value control

Key Features:

- Terminal User Interface (TUI)
- DBC file support
- Individual control of the sending frequency for each message (10ms - 10s)
- Real-time monitoring with automatic decoding
- Single and general stop for all messages



⚠ SocketCAN is required to connect to CAN interfaces. If you don't have CAN hardware, use the virtual interface vcan0.

Supported Operating Systems

- **Linux (recommended):** Native support for SocketCAN. All features (virtual and real) work on Linux.
- **Windows:** Releases include an executable, but SocketCAN requires WSL or an emulator; we recommend WSL2 with an appropriate network configuration.
- **macOS:** A release is provided, but SocketCAN is not native; for testing, we recommend using a Linux or WSL machine.

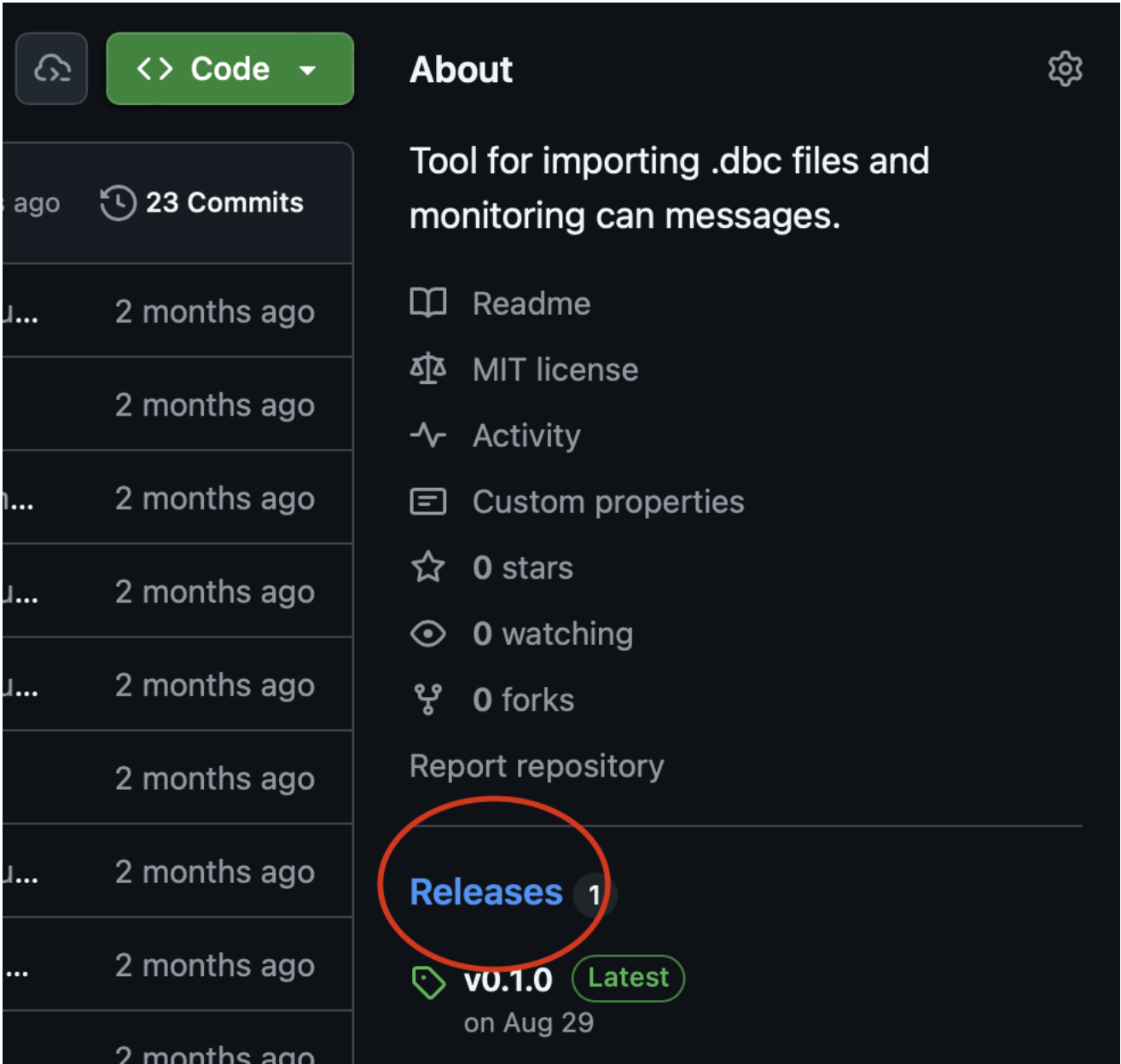
QUICK INSTALLATION



Download from GitHub Releases:

Go to:
Download the file for the appropriate system:

- Linux: `can-debug\_Linux\_x86\_64.tar.gz`
- macOS Intel: `can-debug\_Darwin\_x86\_64.tar.gz`
- macOS Apple Silicon: `can-debug\_Darwin\_arm64.tar.gz`
- Windows: `can-debug\_Windows\_x86\_64.zip`



QUICK INSTALLATION



Setup:

Extract with:

```
tar -xzf can-debug_*.tar.gz  
cd can-debug
```

Make executable (Linux/macOS):

```
chmod +x can-debug
```

BUILD FROM SOURCE



Requirements:

Install Go version 1.19 or higher on Ubuntu/Debian:

```
sudo apt install golang-go
```

Build steps:

```
git clone https://github.com/squadracorsepolito/can-debug.git  
cd can-debug  
go build -o can-debug
```


CAN SETUP, RUNNING THE TOOL



With vcan (quick):

Run the helper script to create a vcan interface and automatically start the application:

```
./TestVcanSetUp.sh
```

With vcan (manual):

```
# create virtual CAN network
sudo modprobe can
sudo modprobe can_raw
sudo modprobe vcan
sudo apt install can-utils # optional but useful (cansend, cangen, candump)
sudo ip link add dev vcan0 type vcan
sudo ip link set up vcan0
# verify:
ip a show vcan0
```



With a real CAN interface:

```
sudo modprobe can
sudo modprobe can-raw
sudo modprobe <appropriate_module>
sudo apt install can-utils      # optional but useful
sudo ip link set <CanNetworkName> type can bitrate <bitrate>
sudo ip link set up <CanNetworkName>
ip a show <CanNetworkName>
```

Replace <CanNetworkName> and <bitrate> with your interface name and bitrate.



After having completed the can network setup, In another terminal you can check the behaviour:

- To observe frames sent by the program (press `Ctrl+C` to stop monitoring):

```
candump vcan0
```

- To send a frame to the program (example):

```
cansend vcan0 123#DEADBEEF
```

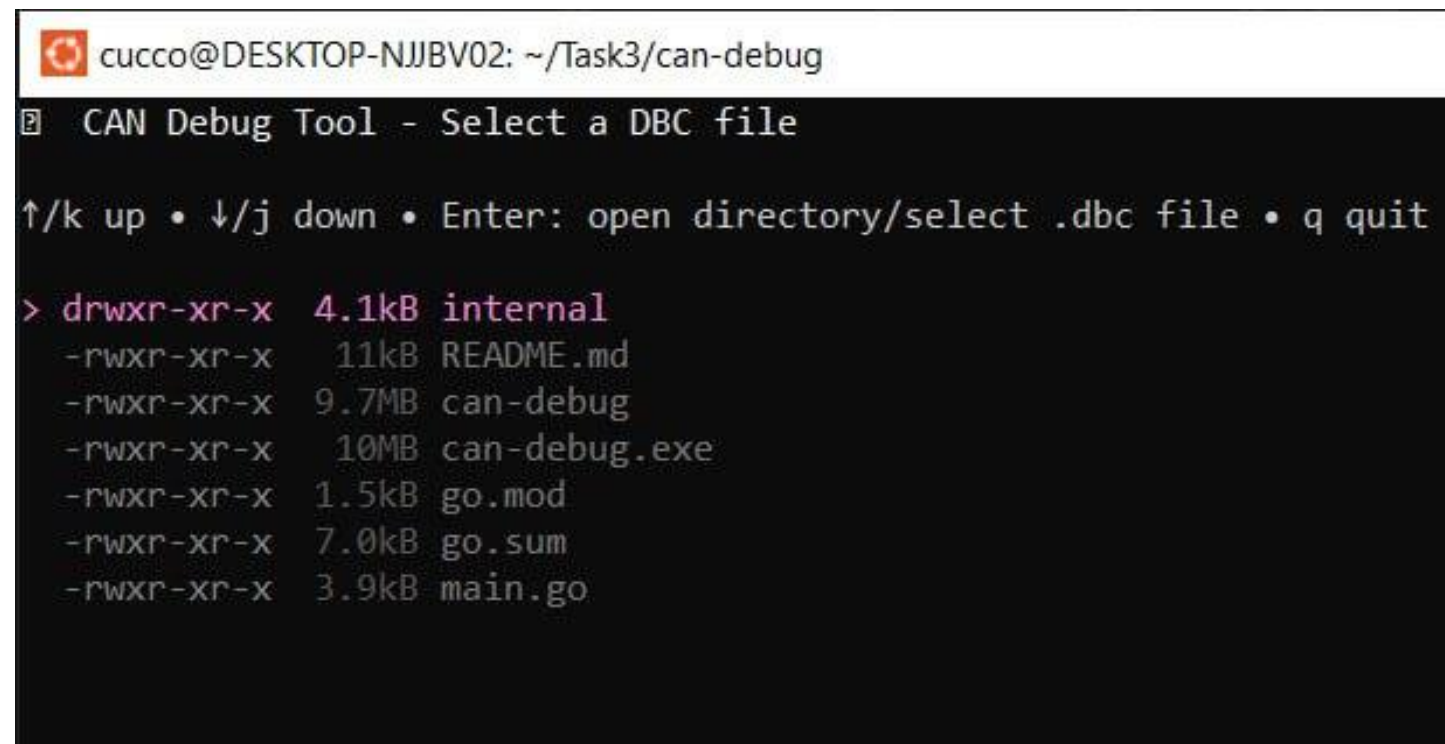
A simple DBC file is available for testing purposes at `internal/test/MCB.dbc`

STARTING MODE



```
./can-debug [canNetworkName] path/to/file.dbc # Directly specify the DBC  
./can-debug [canNetworkName] # Use the file picker
```

```
./can-debug -help # Show help
```



```
cucco@DESKTOP-NJJBV02: ~/Task3/can-debug  
CAN Debug Tool - Select a DBC file  
↑/k up • ↓/j down • Enter: open directory/select .dbc file • q quit  
> drwxr-xr-x 4.1kB internal  
-rwxr-xr-x 11kB README.md  
-rwxr-xr-x 9.7MB can-debug  
-rwxr-xr-x 10MB can-debug.exe  
-rwxr-xr-x 1.5kB go.mod  
-rwxr-xr-x 7.0kB go.sum  
-rwxr-xr-x 3.9kB main.go
```

 **The file picker starts from the current working directory.**

Make sure you're in the correct directory before starting. You can navigate to subdirectories, but don't go "above" the current directory.

DEMO



Selecting the mode:

- Send mode
- Receive mode

It is recommended to open two terminals, one in send mode and one in receive mode.

```
cucco@DESKTOP-NJJBV02: ~/Task3/can-debug
CAN Debug Tool - Choose mode (File: internal/test/MCB.dbc)

Select use mode:

> [ ] Send CAN messages
  [ ] Receive and monitor CAN messages

↑/k up • ↓/j down • Enter confirm • Tab back to file selection • q quit
```

MESSAGE SELECTION



Select the messages you want to send/monitor:

- In **send** mode, the selection is **single**.
- In *receive* mode, the selection is **multiple**.

cucco@DESKTOP-NJJBV02: ~/Task3/can-debug

Select signals:

Navigation: ↑/k up • ↓/j down • / filter • Tab back to mode selection • q quit
Actions: Space select/deselect • Enter start monitoring

IMU__Gyro
ID: 0x38B

IMU__Euler
ID: 0x39B

IMU__GPS1_Position
ID: 0x3AB

IMU__VelocityBody
ID: 0x3BB

IMU__TrackSlipCurv
ID: 0x3CB

NLG5_ST
ID: 0x610

NLG5_ACT_I
ID: 0x611

cucco@DESKTOP-NJJBV02: ~/Task3/can-debug

Select signals:

Navigation: ↑/k up • ↓/j down • / filter • Tab back to mode selection • q quit
Actions: Space select message • Enter configure sending

IMU__Gyro
ID: 0x38B

IMU__Euler
ID: 0x39B

IMU__GPS1_Position
ID: 0x3AB

IMU__VelocityBody
ID: 0x3BB

IMU__TrackSlipCurv
ID: 0x3CB

NLG5_ST
ID: 0x610

NLG5_ACT_I
ID: 0x611



Available controls:

- **Signal values:** change via input field
- **Frequency:** from 10ms to 10s for each message
- **Single shot:** send a single message with the 's' key
- **Cyclic:** continuous sending, configurable
- **Stop All:** stops all messages immediately

```
cucco@DESKTOP-NJJBV02: ~/Task3/can-debug
Configure message values to send (File: internal/test/MCB.dbc)

Navigation: ↑/k up • ↓/j down • Tab back • q quit
Action: Enter send message • Space toggle message • ↔ adjust message cycle • s stop all

Message      ID      Signal      Cycle(ms)  Status      Value
NLG5_TEMP    0x613    NLG5_P_TMP (degC)    1000      [?] stopped  > 197
NLG5_TEMP    0x613    NLG5_TMP_EXT1 (degC) 1000      [?] stopped  > 15.7
NLG5_TEMP    0x613    NLG5_TEMP_EXT2 (degC) 1000      [?] stopped  > -2065.3
NLG5_TEMP    0x613    NLG5_TMP_EXT3 (degC) 1000      [?] stopped  > 1682

Status: [?] Sent message 'NLG5_TEMP' (4 signals) once: [7 178 0 156 175 83 65 180]
```




Monitoring screen features:

- CAN messages decoded in real time
- Signal values updated automatically
- Last update timestamp

```
cucco@DESKTOP-NJJBV02: ~/Task3/can-debug
Real time CAN monitoring | Last update: 17:41:29.879
↑/k up • ↓/j down • Tab back to message selection • q quit

Message      ID      Signal      Value      Raw      Type
IMU__GPS1_Position  0x3AB  LATITUDE (°)  107.8263808  bit 0:31  s32_-214.7483648-21...
IMU__GPS1_Position  0x3AB  LONGITUDE (°)  0          bit 32:63  s32_-214.7483648-21...
IMU__VelocityBody  0x3BB  VELOCITY_X (m.s-1)  [In attesa dati]  bit 0:15  s16_-327.68-327.67_...
IMU__VelocityBody  0x3BB  VELOCITY_Y (m.s-1)  [In attesa dati]  bit 16:31  s16_-327.68-327.67_...
IMU__VelocityBody  0x3BB  VELOCITY_Z (m.s-1)  [In attesa dati]  bit 32:47  s16_-327.68-327.67_...
IMU__Gyro          0x38B  GYRO_X (rad.s-1)  16.386      bit 0:15  s16_-32.768-32.767_...
IMU__Gyro          0x38B  GYRO_Y (rad.s-1)  28.72       bit 16:31  s16_-32.768-32.767_...
IMU__Gyro          0x38B  GYRO_Z (rad.s-1)  0           bit 32:47  s16_-32.768-32.767_...
```